Inkterface Assembly (p. 1/2)

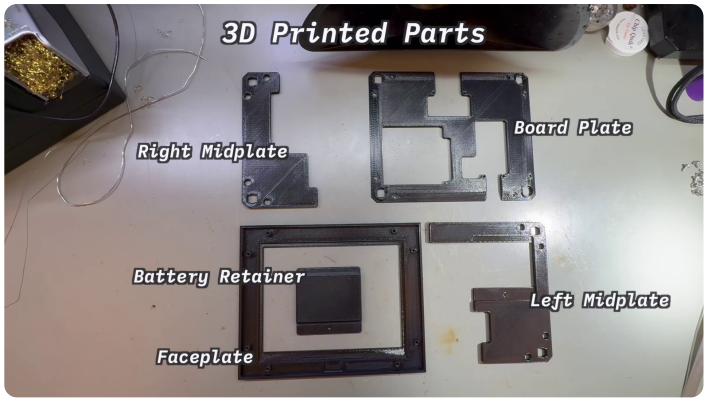
Collect hardware:
> 1x 5.83" eInk Panel
> 1x Adafruit eInk Breakout
> 1x ESP32-S3 Feather w/ 2MB PSRAM
> 1x LP803860 Battery
> 4x K&J Magnetics SB443-OUT Magnets
> 13x M2.6 x 5mm Machine Screws

See project readme for links.



Print plastic parts:
> 1x Faceplate
> 1x Battery Retainer
> 1x Right Midplate
> 1x Left Midplate
> 1x Board Plate

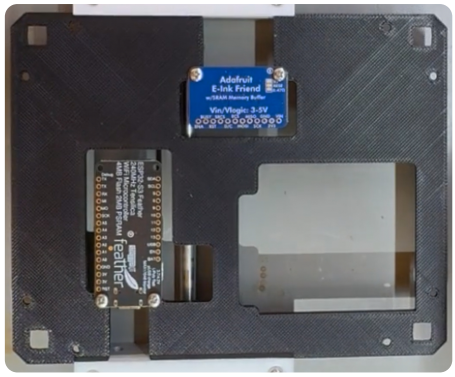
There are STEP and STL files provided in the CAD folder of the project.



Attach boards to board plate:
Use 4 M2.5 x 5mm screws, the breakout and feather.

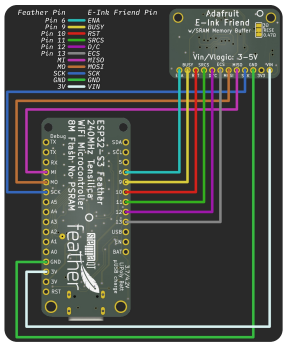
The boards should lay on the flat side, with the connectors and components facing down.

Mounting them like this makes wiring and soldering easier. ***Tighten screws gently to avoid damaging the plastic!***



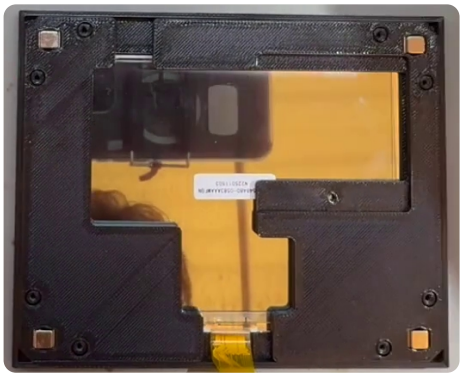
Wire boards together:
Solder fine gauge wire connections between the boards according to the wiring diagram.

I prefer ~30AWG stranded wire with silicone insulation, but just try to keep the wires as short and direct as possible.





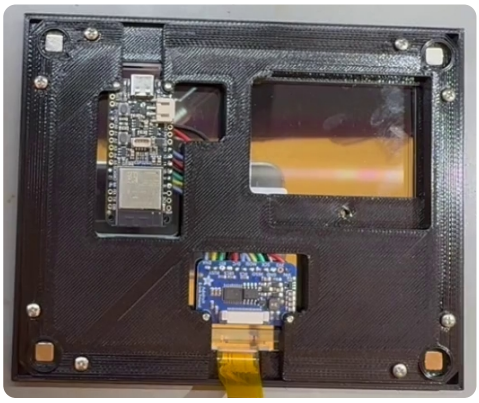
Assemble faceplate and midplates:
Remove protective film from eInk panel.
Place panel in matching recess in faceplate.
Place midplates down as seen in image.
Install 4 K&J SB443-OUT magnets.
  1 in each corner, you may need to press them in forcefully depending on your printer.



Inkterface Assembly (p. 2/2)

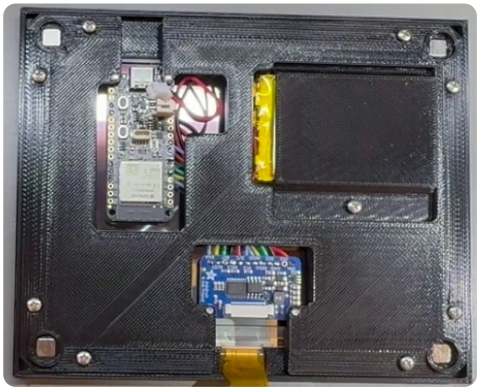
Mount board plate over midplates:
Carefully position wires so that they fit into the channel between the midplates.

Use 8 M2.5 x 5mm screws, gently tightened to avoid damaging the plastic.



Mount battery and battery retainer:
Feed battery cable under board plate and up next to the battery connector on the feather.

Place the battery retainer over the battery and fix it in place with 1 M2.5 x 5mm screw.



Connect flex cable to breakout:
Carefully unlatch the connector and feed the flex cable into it.

The cable should seat as shown in the image.

Carefully latch the connector.



Connect the battery to the feather:
Connect the battery to the feather.

You should see a red light, this indicates that the board has power.



Assembly Complete!
The next step is to flash the firmware, please see the readme for instructions on setup, building, and flashing!